

Software Engineering – Lab 1

Requirements Engineering & UML Use-Case Modelling

Problem Statement #16 – Healthcare & Telemedicine

Remote Patient Vitals Alert & Monitoring App

Student Name: Tushar N

SRN: PES1UG24CS504

1. Problem Overview

The Remote Patient Vitals Alert & Monitoring App is designed to continuously monitor post-operative patients from a remote location. The system receives vital telemetry such as SpO2, heart rate, and blood pressure, checks the readings against configured clinical thresholds, and raises alerts when a critical value is detected. Alerts are then escalated through a caregiver matrix so that the responsible on-call caregiver can respond quickly.

2. Target Actors

- Remote Patient – the patient whose vital readings are being monitored.
- On-Call Caregiver – receives alerts and responds to critical patient conditions.

3. Functional Requirements

The following requirements are based on the given problem statement and follow the Lab 1 format: Req ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall continuously evaluate incoming patient vital readings against configured clinical thresholds and flag critical anomalies within 2 seconds.	High	Pass: A heart rate above 140 BPM triggers an alert within 2 seconds. Fail: A critical reading is ignored or the alert is delayed beyond 5 seconds.	Ensures abnormal patient conditions are detected quickly so that care can be initiated without unnecessary delay.
FR-002	Functional	The system shall continuously receive and record SpO2, heart rate, and blood pressure readings for a monitored patient.	High	Pass: Each valid incoming reading is stored with the patient ID and timestamp. Fail: A valid reading is received but not recorded.	Provides a continuous record of the patient's vital condition and supports reliable monitoring.
FR-003	Functional	The system shall notify the assigned	High	Pass: The assigned caregiver receives	Makes sure the responsible

		on-call caregiver when a patient's vital reading crosses a configured critical threshold.		an alert identifying the patient and the abnormal vital. Fail: No notification is generated after a confirmed threshold breach.	caregiver is informed when immediate attention may be required.
FR-004	Functional	The system shall escalate an unresolved critical alert to the next caregiver in the configured caregiver matrix.	High	Pass: If the first caregiver does not acknowledge a critical alert within the configured response period, the next caregiver is notified. Fail: Escalation does not occur after the response period expires.	Provides a backup path for critical situations when the first caregiver is unavailable.
FR-005	Functional	The system shall allow the on-call caregiver to view the patient's current vital readings and alert status after receiving an alert.	Medium	Pass: The caregiver can open the alert and view the latest available SpO2, heart rate, and blood pressure readings. Fail: Current readings or alert status cannot be displayed.	Gives the caregiver the information needed to understand the alert and decide the next response.

4. Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Availability	The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.	High	Pass: Load testing confirms support for at least 500 simultaneous telemetry streams while maintaining the required service availability. Fail: The system cannot sustain 500 streams or the measured availability is below 99.99%.	Supports reliable continuous monitoring even when many patients are being monitored at the same time.
NFR-002	Security & Privacy	The system shall protect patient telemetry and alert information from unauthorized	High	Pass: An unauthenticated or unauthorized user cannot access patient telemetry or	Patient vital information is sensitive, so access

		access and shall allow only authenticated and authorized users to view patient monitoring data.		alert details. Fail: Restricted patient information is accessible without the required authorization.	must be limited to authorized users.
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5. Core Use-Case Flow Specification

Field	Specification
Use Case Name & ID	UC-01: Monitor Patient Vitals & Generate Alert
Goal	To continuously monitor a patient's vital readings, detect critical abnormalities, generate an emergency alert, and notify the assigned on-call caregiver for timely response.
Primary Actor	Remote Patient
Secondary Actors	On-Call Caregiver, Notification Service
Preconditions	1. Patient is registered in the monitoring system.2. Patient's telemetry stream is active and transmitting vital readings.3. Clinical thresholds and caregiver escalation rules are configured.
Postconditions	1. A critical alert is generated when a vital reading crosses the configured threshold.2. The alert is sent to the assigned on-call caregiver.3. The alert and corresponding vital reading are recorded in the monitoring history.
Main Success Scenario (MSS)	1. System continuously receives the patient's vital readings such as SpO2, heart rate, and blood pressure.2. System validates and records the incoming readings with a timestamp.3. System compares the readings against the configured clinical thresholds.4. System detects a critical anomaly when a vital value crosses its configured threshold.5. System generates an emergency alert containing the patient identifier, abnormal vital value, and time of detection.6. System identifies the responsible on-call caregiver from the configured caregiver matrix.7. System sends the emergency alert to the assigned caregiver.8. Caregiver views the patient's latest vital readings and alert details.9. Caregiver acknowledges the alert and the system records the acknowledgement.

Field	Specification
Alternate Flows	4a. Normal Vital Reading:1. System determines that all vital readings are within the configured safe thresholds.2. No emergency alert is generated.3. System continues monitoring incoming readings.7a. Alert Not Acknowledged:1. The assigned caregiver does not acknowledge the critical alert within the configured response period.2. System marks the alert for escalation.3. System selects the next caregiver according to the caregiver matrix.4. System sends the escalated alert to the next caregiver.2a. Invalid or Missing Telemetry:1. System detects missing or invalid vital data.2. System records the telemetry issue and continues attempting to receive valid readings.3. System notifies the appropriate caregiver if the telemetry interruption exceeds the configured limit.
Special Requirements	The system shall detect and flag critical vital anomalies within 2 seconds of receiving the corresponding telemetry reading. The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.

6. UML Relationship Specification

- «include»: UC-01 Monitor Patient Vitals includes UC-02 Detect Critical Anomaly because checking the received readings against configured thresholds is a required part of continuous monitoring.
- «include»: UC-03 Send Emergency Alert includes UC-06 Acknowledge Alert because alert handling requires the caregiver to acknowledge the notification.
- «extend»: UC-04 Escalate Alert extends UC-03 Send Emergency Alert when the alert remains unacknowledged for the configured response period.

For the UML diagram, place the Remote Patient and On-Call Caregiver outside the system boundary. Place the core use cases inside the boundary and connect each actor to the relevant use cases. The required «include» and «extend» relationships should be shown using dashed dependency arrows.

7. Core Use-Case Flow Specification

Use Case: UC-03 – Send Emergency Alert

Goal: Notify the responsible on-call caregiver when a critical patient vital anomaly is detected.

Preconditions

1. The patient is registered in the monitoring system.
2. The patient's telemetry stream is active.
3. Clinical thresholds and the caregiver escalation matrix are configured.

4. An on-call caregiver is available in the system.

Postconditions

5. A critical alert is created and associated with the patient.
6. The assigned on-call caregiver is notified.
7. The alert remains available for acknowledgement and further escalation if required.

Main Success Scenario

8. The system receives a patient's latest vital readings.
9. The system compares the readings with the configured clinical thresholds.
10. The system detects a critical anomaly, such as a heart rate above the configured limit.
11. The system creates an emergency alert containing the patient identifier, abnormal vital, reading, and time.
12. The system identifies the responsible on-call caregiver from the caregiver matrix.
13. The system sends the emergency alert to the caregiver.
14. The caregiver receives the alert and opens the patient's monitoring information.
15. The caregiver acknowledges the alert and begins the required response.
16. The system records the alert acknowledgement and keeps the event in the monitoring history.

Alternate Flow – Alert Not Acknowledged

17. The system sends the critical alert to the assigned on-call caregiver.
18. The configured response period expires without acknowledgement.
19. The system marks the alert as requiring escalation.
20. The system selects the next caregiver in the escalation matrix.
21. The system sends the escalated alert to the next caregiver and records the escalation event.

8. Visual UML Structural Representation:

